

all this.

$m, n > 0$ 2k.

Preview

mid
school

$$a^m \cdot a^n = a^{m+n}$$

$$(a^m)^n = a^{mn}$$

$$a^m \div a^n = \begin{cases} \cdot a^{m-n} & (m > n, a \neq 0) \\ \cdot 1 & (m = n, a \neq 0) \\ \cdot \frac{1}{a^{n-m}} & (m < n, a \neq 0) \end{cases}$$

$$(ab)^n = a^n b^n.$$

• When 2k.

-1) $a \neq 0$, all 2k 2k.

$$\cdot a^0 = 1$$

$$\cdot a^{-n} = \frac{1}{a^n}$$

-2) $a \neq 0, b \neq 0$ AND m, n 2k

$$\cdot a^m \cdot a^n = a^{m+n}$$

$$\cdot a^m \div a^n = a^{m-n}$$

$$\cdot (a^m)^n = a^{mn}.$$

$$\cdot (ab)^n = a^n b^n.$$

• When 2k.

-1) 2k 2k

$a > 0$ AND $m, n (n \geq 2)$ 2k.

$$\cdot a^{\frac{1}{n}} = \sqrt[n]{a}$$

$$\cdot a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

-2) law.

$a > 0, b > 0$ AND r, s 2k.

$$\cdot a^r \cdot a^s = a^{r+s}$$

$$\cdot a^r \div a^s = a^{r-s}$$

$$\cdot (a^r)^s = a^{rs}$$

$$\cdot (ab)^r = a^r \cdot b^r.$$